

Keywords: large-scale training • mixture-of-experts • Ascend NPU • post-training • operations research AI

SLAI T-Rex: Full-Parameter Post-training of the DeepSeek-V4 Family on Ascend SuperPOD

Huawei Cloud Achieves a $2.93\times$ MFU Improvement Over the GPU Baseline for Trillion-Parameter MoE Post-Training on Ascend NPUs, Demonstrating Hardware-Agnostic Frontier LLM Training

By SLAI Team (Huawei Cloud AI Lab)

arXiv:2607.20145 • July 2026

Training frontier large language models has been synonymous with NVIDIA GPU clusters. A new technical report from Huawei Cloud challenges this assumption: SLAI T-Rex demonstrates that full-parameter post-training of DeepSeek-V4—a 671-billion-parameter Mixture-of-Experts model—can be performed efficiently on Ascend NPU SuperPOD clusters via a hierarchical optimization framework spanning model parallelism, communication scheduling, and low-level kernel engineering. The system achieves 34.22% Model FLOPs Utilization (MFU)—a $2.93\times$ improvement over the open-source GPU baseline—and produces an operations-research specialist model that outperforms GPT-5.4-Mini.

AT A GLANCE

Problem: Full-parameter post-training of trillion-parameter MoE models on non-GPU hardware suffers from memory pressure, communication overhead, and kernel inefficiency.

Why it matters: Hardware diversity in large-scale LLM training reduces dependency on a single chip vendor and expands the ecosystem of frontier model development.

Method: Expert Shard Parallelism, communication-aware scheduling, and Ascend-specific CANN kernel optimizations, combined into a three-level hierarchical framework.

Result: 34.22% MFU ($2.93\times$ over baseline); OR-specialist model achieves 71.81% zero-shot Pass@1, +3.98 pp over GPT-5.4-Mini.

Evidence: Validated on DeepSeek-V4-Flash (671B active params) across a full CPT+SFT post-training workflow.

The Big Picture

DeepSeek-V4 is a Mixture-of-Experts model with 671 billion total parameters (37 billion active per token), trained with 4096 experts grouped into routed and shared expert layers. Post-training this model from scratch on Ascend hardware requires solving problems at three intersecting layers: how to distribute the model’s enormous weight across memory-constrained NPU nodes, how to orchestrate the expert-parallel all-to-all communications that differ fundamentally from NVLink-based topologies, and how to write efficient NPU kernels for the specific operations (gated attention, top- k routing, grouped GEMM) that account for the bulk of compute time.

Why Existing Approaches Fall Short

The open-source DeepSeek-V4 training recipe was designed for NVIDIA GPU clusters with NVLink interconnects. Porting it to Ascend SuperPOD directly yields only $\sim 11.7\%$ MFU because:

- Ascend’s ScaleUp network has different bandwidth/latency characteristics than NVLink, making the GPU pipeline schedule suboptimal.
- Expert parallelism requires all-to-all communication patterns that expose the Ascend cluster’s inter-pod bandwidth as a bottleneck.
- Standard CUDA-derived kernels are mapped inefficiently to Ascend’s cube tile architecture, leaving the NPU’s SIMD units underutilized.

Core Mechanism

SLAI T-Rex addresses these challenges through a **three-level hierarchical optimization framework**:

Level 1 — Model Parallelism: Expert Shard Parallelism (ESP) shards each expert’s weight matrix across multiple NPU pods, reducing per-device peak memory. A modified pipeline schedule overlaps expert computation with inter-pod communication by reordering micro-batches to ensure that communication and compute streams are never both idle.

Level 2 — Communication Orchestration: A communication-aware scheduler predicts expert load imbalance from recent routing statistics, adjusts micro-batch boundaries to reduce hot-expert serialization, and overlaps all-to-all token dispatches with local attention computations via asynchronous execution streams.

Level 3 — Kernel Engineering: Custom CANN kernels implement fused gating (softmax + top- k selection), Flash Attention adapted to Ascend’s HBM tile layout, and gradient all-reduce with dynamic bucketing tuned to Ascend’s memory bandwidth profile.

Technical Formulation

For an MoE layer with E experts and top- k routing, the routing gate selects:

$$g(x) = \text{TopK}(\text{softmax}(W_g x), k)$$

and the layer output is:

$$\text{MoE}(x) = \sum_{e \in \text{TopK}(x)} g_e(x) \cdot \text{FFN}_e(x)$$

Expert parallel training communicates $O(B \cdot k \cdot d)$ token embeddings per step (where B is batch size and d is hidden dimension) across all expert pods. The SLAI framework fuses the dispatch (input routing) and combine (output aggregation) all-to-all operations and prefetches expert weights from HBM, reducing effective communication latency by 38% compared to the baseline.

MFU is defined as the ratio of observed compute to peak theoretical compute:

$$\text{MFU} = \frac{\text{Observed TFLOPs/sec}}{\text{Peak Theoretical TFLOPs/sec (Ascend cluster)}}$$

The SLAI T-Rex system achieves $\text{MFU} = 34.22\%$ on DeepSeek-V4-Flash post-training, compared to 11.68% for the open-source GPU recipe adapted naively to Ascend.

Learning or Inference Procedure

The OR-specialist model development follows a two-stage workflow:

1. **Continued Pretraining (CPT):** DeepSeek-V4-Flash is continued-pretrained on a curated corpus of operations research literature, textbooks, LP/MILP problem libraries, and synthetic problem-solution pairs.
2. **Supervised Fine-tuning (SFT):** 10,000 high-quality SFT samples spanning four task categories (LP, MILP, scheduling, routing) and three problem representations (mathematical, natural language, structured JSON) are used to align the model’s output format and reasoning style.

The final model is evaluated zero-shot at Pass@1 without any inference-time search.

What the Guarantee Says

The paper does not provide theoretical guarantees on MFU or task accuracy. The MFU improvement is empirically validated on a full-scale DeepSeek-V4-Flash post-training run, and the $2.93\times$ figure represents end-to-end wall-clock throughput improvement including all communication and memory overhead.

Experimental Findings

Model	MFU	OR Pass@1
GPU baseline (open-source)	11.68%	—
SLAI T-Rex	34.22%	—
DeepSeek-V4-Flash (base)	—	60.54%
GPT-5.4-Mini	—	67.83%
SLAI T-Rex OR-specialist	—	71.81%

The $2.93\times$ MFU improvement translates directly to proportionally reduced wall-clock training time and therefore training cost on the same hardware budget. The OR-specialist achieves highest zero-shot accuracy among evaluated models on a held-out OR benchmark spanning diverse problem types and sizes.

Ablations and Interpretation

ESP alone vs. full framework: Expert Shard Parallelism alone raises MFU from 11.68% to 19.4%; adding the communication scheduler raises it to 27.1%; adding Ascend-specific kernels reaches 34.22%.

CPT vs. SFT-only: The OR-specialist without CPT (SFT-only on the base model) achieves 66.3% Pass@1, 5.5 pp below the full CPT+SFT model, confirming that domain vocabulary and mathematical notation pretraining is material.

Data representation: Models trained on all three problem representations outperform single-format training by 2.1–3.8 pp, suggesting that diverse input format exposure improves generalization.

Reference

SLAI Team. **SLAI T-Rex: Full-Parameter Post-training of the DeepSeek-V4 Family on Ascend SuperPOD**. arXiv:2607.20145, July 2026. <https://arxiv.org/abs/2607.20145>